SACHIN ABEYWICKRAMA

Software Engineer • Full-Stack Developer • AI & Machine Learning

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Sachin Abeywickrama

Software Engineering graduate with one year of industry experience developing enterprise web applications and AI-driven software solutions. Experienced in Java, Spring Boot, Python, React, Angular, TypeScript, REST APIs, and relational databases through both industry and academic projects.
Passionate about designing scalable software, building intelligent applications, and solving complex real-world problems using machine learning. Seeking Graduate Software Engineer, Associate Software Engineer, or Software Engineering Internship opportunities where I can continue learning from experienced engineers while delivering high-quality software solutions.

TECHNICAL SKILLS

Programming Languages
- Java
- Python
- JavaScript
- TypeScript

Frontend
- React
- Next.js
- Angular
- HTML5
- CSS3
- SCSS
- Tailwind CSS

Backend
- Spring Boot
- FastAPI
- Node.js
- REST APIs

Machine Learning
- PyTorch
- Deep Learning
- Time-Series Forecasting
- Explainable AI
- Ensemble Learning
- Data Preprocessing
- Model Evaluation
- Backtesting

Databases
- MySQL
- MongoDB
- Firebase

Tools & Platforms
- Git
- GitHub
- IntelliJ IDEA
- VS Code
- Postman
- Linux
- Kali Linux

Cybersecurity
- OWASP Fundamentals
- Secure Authentication
- Network Security Concepts

PROFESSIONAL EXPERIENCE

SDB Bank **June 2024 - June 2025**
Software Engineering Intern

- Contributed to the development and maintenance of enterprise applications, including the Asset Tracking System and the Bank Intranet.
- Developed backend REST APIs using Spring Boot and integrated them with Angular frontend components.
- Built responsive UI components using Angular, TypeScript, and SCSS.
- Designed and maintained MySQL databases supporting internal business applications.
- Implemented secure authentication and role-based access control features.
- Collaborated with senior engineers throughout requirements gathering, UI design, debugging, testing,
- and deployment.
- Used Git and GitHub for version control and collaborative software development.
- Assisted with production deployments, post-deployment validation, user feedback collection, and issue resolution.

KEY PROJECTS

AquaAdapt: Water Quality Prediction System for Aquaculture Farmers
(Python, FastAPI, Next.js, React, PyTorch, TypeScript, Tailwind CSS, Firebase, Open-Meteo API)
Final Year Research Project

- Designed and developed a full-stack aquaculture monitoring platform for fish farmers.
- Built deep learning models using PyTorch to predict future water quality parameters including pH, temperature, and turbidity.
- Integrated weather forecasting data through the Open-Meteo API.
- Developed an interactive dashboard displaying predictions, alerts, recommendations, and explainable AI insights.
- Implemented model validation and forecasting features to improve aquaculture decision-making.

GestureSway: Hand Gesture Controlled Snake Game
(Python, React, Node.js, MongoDB)

- Developed a machine learning-powered browser game controlled entirely through hand gestures.
- Applied computer vision techniques to improve accessibility for users with limited hand mobility.
- Built the application using the MERN stack.

Thread-Safe Hospital Management System
(Java, Concurrent Programming)

- Developed a concurrent hospital management system demonstrating thread synchronisation and resource sharing.
- Applied Java concurrency concepts to simulate safe multi-user hospital operations.
- Focused on thread safety, synchronisation, and efficient resource management.

Westminster Shopping Manager
(Java, Java Swing)

- Designed and developed a desktop shopping application with a graphical user interface.
- Implemented product management, customer purchasing, and inventory features using object-oriented programming principles.

EDUCATION

University of Westminster **September 2022 - May 2026**

Bachelor of Engineering (Honours) - Software Engineering

Final Year Research: An IoT-Enabled Machine Learning-Based Water Quality Monitoring, Forecasting, and Decision Support System for Aquaculture

Supervisor: Imesh Pathirana

Key Modules: Object Oriented Programming, Concurrent Programming, Database Systems, Algorithms, Server-side Web Development, Operational Research Methods, Web Designing, Cybersecurity

Professional Memberships: IEEE Club, Rotaract Club, IEEE Computer Society

Royal College, Monaragala **2012 - 2021**

- Passed GCE Advanced Levels in 2021 in the Mathematics Stream (Mathematics, Physics and ICT)
- Passed GCE Ordinary Levels in 2018

Leadership & Extracurricular Activities: President of the Literary Society, Senior Prefect of Royal College Monaragala, Member of the Interact Club, and Member of the Badminton Team